

30-Day Roadmap: Cloud & AI Security Engineer

Built on your existing skills: Azure Cloud, Application Security, AI Security, MS SQL **Time commitment:** 2 hrs/day | **Level:** Intermediate → Advanced **Format per day:** ~40 min video, ~30 min reading, ~50 min hands-on

🎯 Final Outcome (Day 30)

A portfolio-ready "Secure AI Application on Azure" project containing:

- A threat-modeled, deployed Azure application with a SQL backend
- Integrated AppSec controls (SAST/DAST, secrets management, IAM hardening)
- An AI/LLM security layer (prompt-injection defenses, PII filtering, model access controls)
- A written security assessment report + architecture diagram
- A polished GitHub repo + LinkedIn write-up you can show interviewers

Week 1 (Days 1-7): Azure Security Foundations Refresh

Milestone: Rebuild your Azure environment with security-first architecture (Zero Trust basics, IAM, networking).

Day	Task	Resource
1	Refresh Azure Zero Trust & Defender for Cloud concepts	Microsoft Learn: "Zero Trust" module + Defender for Cloud docs
2	Set up Azure free tier; deploy a basic web app + SQL DB	Azure Portal + Azure CLI
3	Harden IAM: RBAC, Conditional Access, Managed Identities	Microsoft Learn "Secure Azure resources with RBAC"
4	Network security: NSGs, Private Endpoints, Key Vault	MS Learn "Azure network security" path
5	Enable logging: Azure Monitor + Sentinel basics (free trial)	YouTube: "Azure Sentinel for Beginners" (John Savill)
6	Read: Azure Security Benchmark (skim relevant controls)	Microsoft Cloud Security Benchmark doc
7	Project Checkpoint: Document your hardened architecture (diagram + notes)	Draw.io / Excalidraw

Resources this week: Microsoft Learn (free), John Savill's Technical Training (YouTube), Azure Security Benchmark docs.

Week 2 (Days 8-14): Advanced Application Security

Milestone: Integrate security testing into your app's SDLC (shift-left security).

Day	Task	Resource
8	Review OWASP Top 10 (2021/latest) with focus on injection & auth flaws	owasp.org/Top10
9	Set up SAST scanning (e.g., Semgrep or SonarQube Community) on your app	Semgrep docs / free CLI
10	Set up DAST scanning (OWASP ZAP) against your deployed app	OWASP ZAP official getting-started guide
11	Secure your SQL layer: parameterized queries, least-privilege DB accounts	MS SQL security docs (Microsoft Learn)
12	Secrets management: move credentials to Azure Key Vault	Microsoft Learn "Key Vault fundamentals"
13	Read: "Secure DevOps" / DevSecOps pipeline concepts	freeCodeCamp article on DevSecOps
14	Project Checkpoint: Add a CI/CD pipeline (GitHub Actions) with SAST/DAST gates	GitHub Actions docs

Resources this week: OWASP.org, Semgrep, OWASP ZAP, GitHub Actions docs.

Week 3 (Days 15-21): AI & LLM Security Specialization

Milestone: Add an AI-powered feature to your app and secure it end-to-end.

Day	Task	Resource
15	Study OWASP Top 10 for LLM Applications	owasp.org/www-project-top-10-for-large-language-model-applications
16	Learn prompt injection & jailbreak patterns (conceptual, defensive focus)	YouTube: "Prompt Injection Explained" (search recent talks, e.g. DEF CON AI Village)
17	Build a simple LLM-integrated feature (e.g., chatbot via API) into your app	Anthropic/OpenAI API quickstart docs
18	Implement input/output filtering, PII redaction, rate limiting	Read: Microsoft "Responsible AI" + Azure AI Content Safety docs
19	Explore model access control: API key rotation, least-privilege model access	Azure AI Studio security docs
20	Read case studies of real LLM security incidents (data leakage, injection)	Search recent security blogs (e.g., Embrace The Red, NCC Group AI research)
21	Project Checkpoint: Write a mini threat model for your AI feature (STRIDE-based)	Microsoft Threat Modeling Tool (free)

Resources this week: OWASP LLM Top 10, Azure AI Content Safety docs, Microsoft Threat Modeling Tool.

Week 4 (Days 22-30): Integration, Portfolio & Job Readiness

Milestone: Ship the final project and prepare interview-ready materials.

Day	Task	Resource
22	Combine all pieces: hardened Azure infra + secure app + AI feature	Your own repo
23	Run a full security review (SAST + DAST + manual checklist)	OWASP ASVS checklist (use relevant sections)
24	Write your Security Assessment Report (findings, fixes, residual risk)	Use a simple pentest report template (search "OWASP sample report")
25	Polish architecture diagram + README for your GitHub repo	Draw.io / Excalidraw
26	Record a 3-5 min project walkthrough video (great for LinkedIn/portfolio)	Loom / OBS Studio (free)
27	Update resume & LinkedIn with new project + AI security keywords	—
28	Mock interview prep: cloud security + AI security scenario questions	Search "Azure security interview questions" + "LLM security interview questions"
29	Review weak areas from the month; revisit any skipped topics	Your own notes
30	Final Project Presentation: Publish repo, write-up, and share your walkthrough	GitHub + LinkedIn

Weekly Milestone Summary

Week	Milestone
1	Secure Azure environment deployed and documented
2	AppSec testing integrated into CI/CD pipeline
3	AI feature built and threat-modeled
4	Full project shipped + portfolio/interview materials ready

Tools You'll Use (all free-tier friendly)

- Azure Free Account, Azure CLI, Azure Key Vault, Defender for Cloud, Sentinel (trial)
- Semgrep, OWASP ZAP, GitHub Actions
- Microsoft Threat Modeling Tool, OWASP ASVS

- An LLM API (Anthropic/OpenAI) for the AI feature

Tips

- If a day's task overruns, cut reading time first — hands-on project work is highest priority for interviews.
- Keep a running "learnings log" — it becomes your interview talking points and LinkedIn content.
- Don't aim for a perfect project; aim for a **complete, explainable** one.